
Your solutions need to be handwritten. After writing down your solutions, scan and compile those into a single pdf file. Then submit in the following link:

<https://forms.gle/zmeVaG8jAxVk3x3H8>

[The form will be accessible only from your G-Suite email account.]

Problem 1

Write down regular expressions for the following languages.

- a) $\{w \in \{0, 1\}^* : w \text{ does not end in } 101\}$ (2 points)
- b) $\{w \in \{a, b, c\}^* : w \text{ starts with } abba \text{ and ends in } bac\}$ (2 points)
- c) $\{w \in \{0, 1\}^* : w \text{ has length at least } 3 \text{ and its third symbol is a } 0\}$ (2 points)
- d) $\{w \in \{0, 1\}^* : w \text{ contains an even number of } 0\text{'s or exactly two } 1\text{'s}\}$ (2 points)
- e) $\{w \in \{a, b\}^* : \text{the number of } a\text{'s in } w \text{ is } 1 \text{ more than a multiple of } 3\}$ (2 points)

Problem 2

Describe the languages that the following regular expressions generate. (5 points)

- a) $01(1 + 0)^*$
- b) $0(0 + 1)^* + (0 + 1)^*1$
- c) $(1 + 01)^*(0 + \epsilon)$
- d) $(0 + \epsilon)(10)^*(1 + \epsilon)$
- e) $(0 + 10)^*(1 + \epsilon) + (1 + 01)^*(0 + \epsilon)$

Problem 3

Construct deterministic finite automata for the following languages.

- a) $\{w \in \{0, 1\}^* : w \text{ contains exactly two } 0\text{'s}\}$ (2 points)
- b) $\{w \in \{0, 1\}^* : w \text{ contains an even number of } 0\text{'s and an odd number of } 1\text{'s}\}$ (2 points)
- c) $\{w \in \{0, 1\}^* : \text{the length of } w \text{ is even and } w \text{ contains } 0\text{'s at all the odd positions}\}$ (3 points)
- d) $\{w \in \{0, 1\}^* : w \text{ contains } 01^m0 \text{ as a substring where } m \text{ is divisible by } 3\}$ (3 points)
- e) $\{w \in \{0, 1\}^* : w \text{ contains } 0\text{'s and } 1\text{'s in alternate}\}$ (3 points)
- f) $\{w \in \{0, 1\}^* : \text{the last two symbols of } w \text{ are the same}\}$ (3 points)

- g) $\{ w \in \{0, 1\}^* : w = 0^m 1^n \text{ where } m \text{ is even and } n \geq 0 \}$ (3 points)
- h) $\{ w \in \{0, 1\}^* : w \text{ doesn't contain } 00 \text{ and } 11 \}$ (3 points)
- i) $\{ w \in \{0, 1, 2\}^* : w \text{ when interpreted in three base number system is divisible by nine} \}$ [Try using 3 states maximum] (3 points)

j) [Bonus Question] Let $\Sigma = \{0, 1\}$. For $w \in \Sigma^*$, we denote by $h(w)$ the number of 1s in w . For example, $h(100101) = 3$.

Consider the following language over Σ . As always, $|w|$ is the length of the string w .

$$L = \{ w : |w|^2 + h(w)^2 \text{ is two more than a multiple of three} \}$$

Give the state diagram for a DFA that recognizes L .

Problem 4

Answer the following questions. Add explanations if needed. (5 points)

- a) Are regular expressions $(ba)^+(a^*b^* \mid a^*)$ and $(ba)^*ba^+b^*$ the same?
- b) Write down the shortest nonempty string generated by the expression:
 $(0+1)^* (0 (0+1) 0 + 1 (100 + 001 + 11^*)) 00$
- c) What is the shortest string generated by the regular expression
 $(00 + 01) (101)^* ((00)^* + 110)$ with exactly three 0s?
- d) Suppose, you are given a regular expression: $a(a^*b^* \mid b^+)b$. Now, is the expression $a^*(ab^+)$ equivalent to your given expression?
- e) Are regular expressions $a^*(b \mid b^* \mid b^+)$ and $a^?(a^+b^+ \mid b^* \mid \epsilon)$ the same?

CSE331 Section 07 & 08 Assignment 2

Spring 2025

Total Marks: 40

Due: 08/05/2025, 11:59 PM

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<https://forms.gle/gm6KEaoetcSapHfT9>

Question 1

[10 Points]

Use the Pumping Lemma to demonstrate that the following languages are not regular.

- a) $L1 = \{w \in \{0, 1\}^*: 0^i 1^j \text{ where } i \leq j\}$
- b) $L2 = \{w \in \{a, b, c\}^*: a^i b^j c^{k+2} \text{ where } i = k \text{ and } i, j, k \geq 0\}$
- c) $L3 = \{w1\#w2 \text{ such that } |w1| = 2 \cdot |w2|, \text{ where } w1, w2 \in \{0, 1\}^* \}$
- d) $L4 = \{w \in \{a\}^*: a^{2^n} \text{ where } n \geq 0\}$

Question 2

[20 Points]

Give a context-free grammar for each of the following languages.

Consider, $\Sigma = \{0, 1\}$.

- A. The language of strings that start with 1
- B. The language of strings of the form ww^R
- C. The language of strings that contain the substring 001
- D. The language $\{0^n 10^n \text{ where } n \geq 0\}$
- E. The language $\{0^i 1^j \text{ where } i \leq j\}$
- F. The language $\{1^i 01^j 01^{i+j}\}$
- G. The language $\{0^{3n} w 1^{2n} \text{ where } n \geq 0 \text{ and } w \text{ is a string that contains at least two } 0\text{'s}\}$
- H. The language $\{0^i 10^j \mid i \text{ is a multiple of three where } i, j \geq 0\}$
- I. The language $\{0^i 10^j \mid j = 2 + 3i \text{ where } i, j \geq 0\}$
- J. The language $\{1^i 01^j \mid j \text{ is a multiple of four or } i = 3 + 2j \text{ where } i, j \geq 0\}$

Question 3

[10 Points]

1. Suppose, you are given the following grammar. $\Sigma = \{0, 1\}$

$S \rightarrow SS \mid 0S1 \mid 1S0 \mid 01 \mid 10$

- a) Show that the grammar is ambiguous by demonstrating two different parse trees. [Hint: String 011010]
- b) Find two strings of length six in the grammar above where exactly one parse tree is possible.

2. Suppose, you are given the following grammar. $\Sigma=\{a,b\}$

$$S \rightarrow PE$$
$$P \rightarrow aPa \mid bPb \mid a \mid b$$
$$E \rightarrow aaE \mid abE \mid baE \mid bbE \mid \varepsilon$$

- a) Give a leftmost derivation for the string abababb and sketch the parse tree corresponding to that derivation.
- b) Demonstrate that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (a)) for the same string abababb.

Problem 1

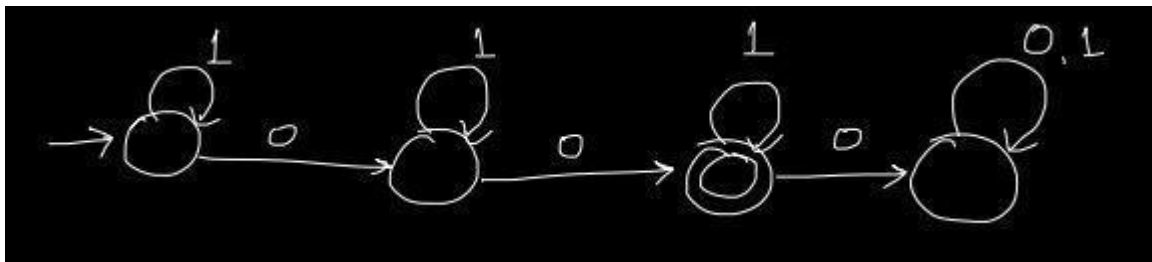
- a) $(0 \mid 1)^* (0 \mid 11 \mid 001) \mid \epsilon \mid 1 \mid 01$
- b) $abba (a|b|c)^* bac \mid abba$
- c) $(0 \mid 1) (0 \mid 1) 0 (0 \mid 1)^*$
- d) $(1^* 0 1^* 0 1^*)^* \mid 0^* 1 0^* 1 0^*$
- e) $(b^* ab^* ab^* ab^*)^* b^* ab^*$

Problem 2

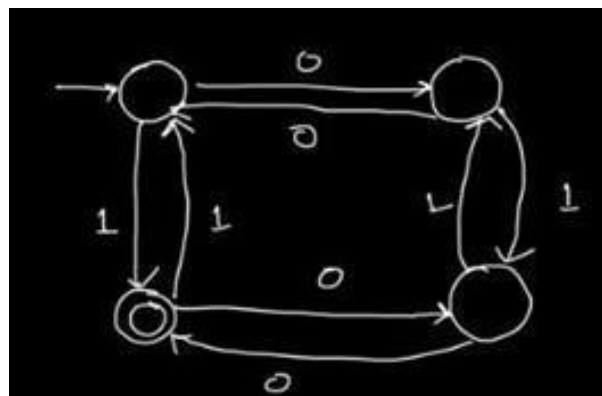
- a) Strings over $\{0,1\}$ that start with 01
- b) Strings over $\{0,1\}$ that start with 0 or end with 1
- c) Strings over $\{0,1\}$ that do not contain 00 as a substring
- d) Strings over $\{0,1\}$ that have neither consecutive 0's, nor consecutive 1's
- e) Strings over $\{0,1\}$ that may have consecutive 0's, or consecutive 1's, but not both

Problem 3

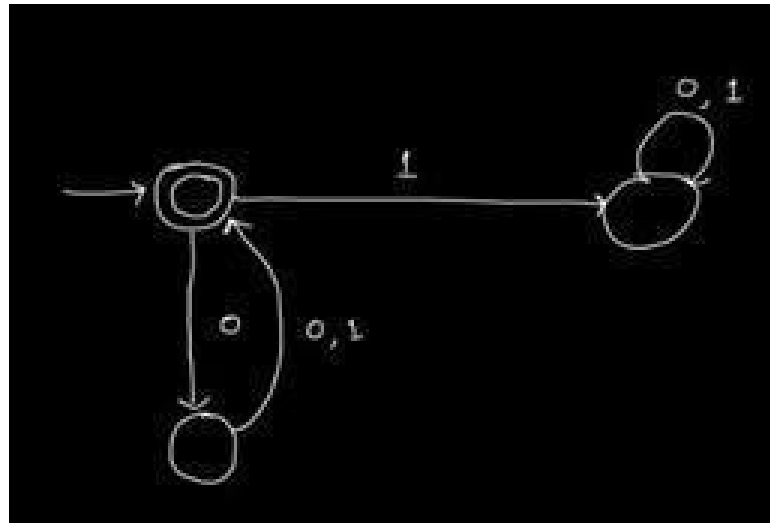
a)



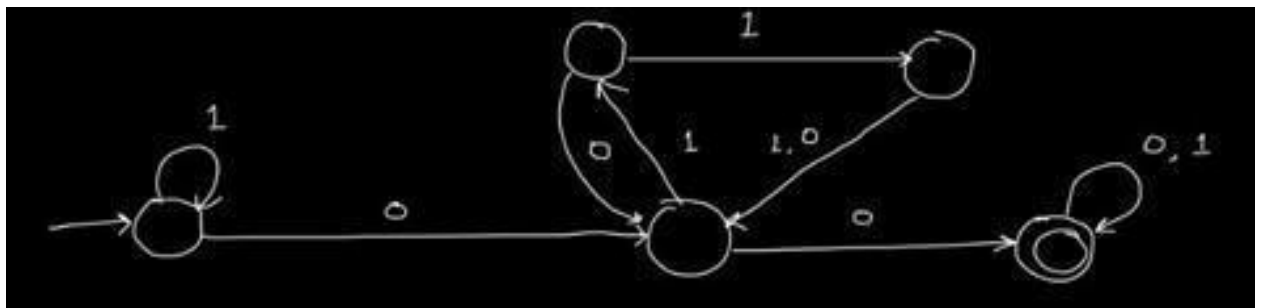
b)



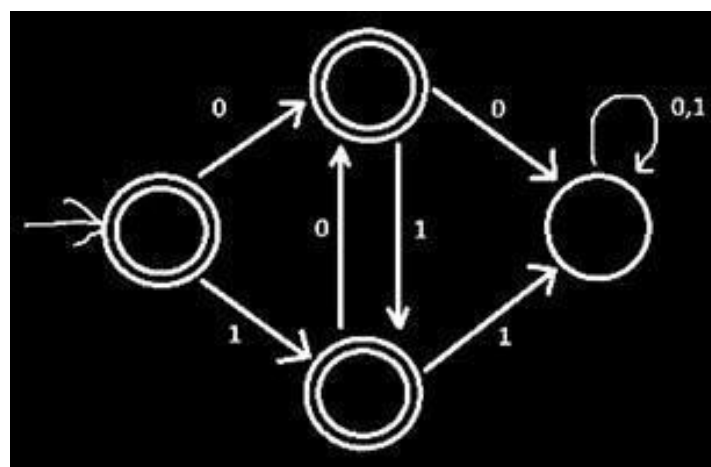
c)



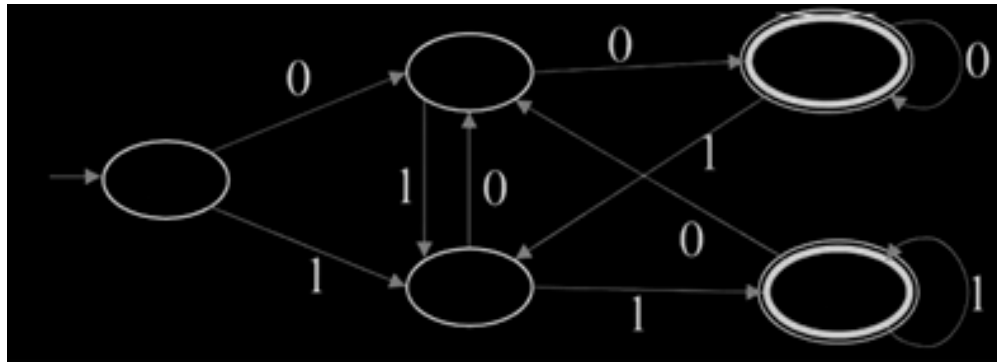
d)



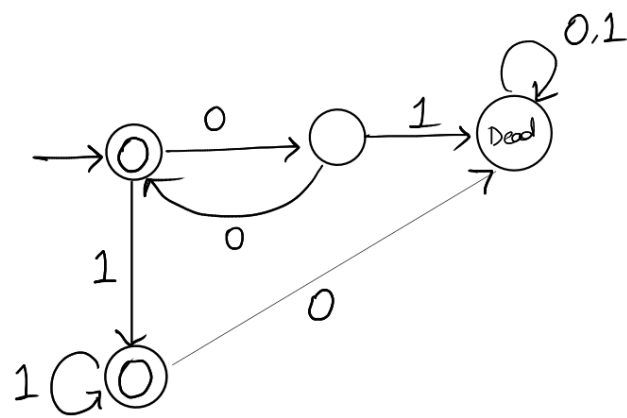
e)



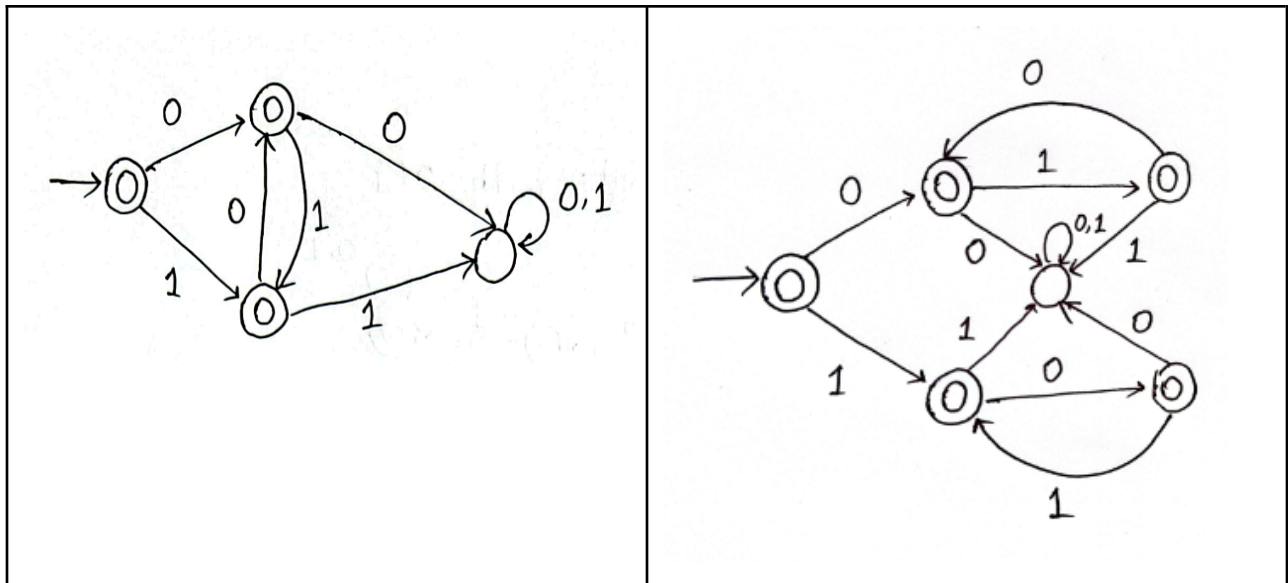
f)



g)

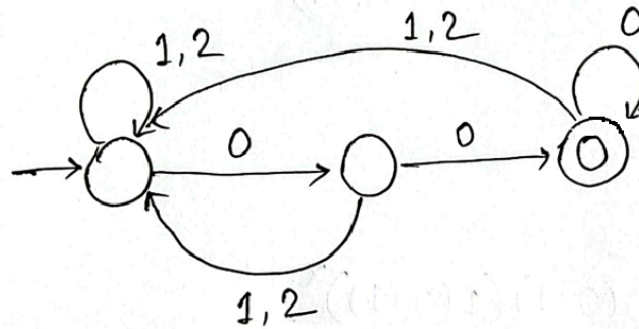


h)



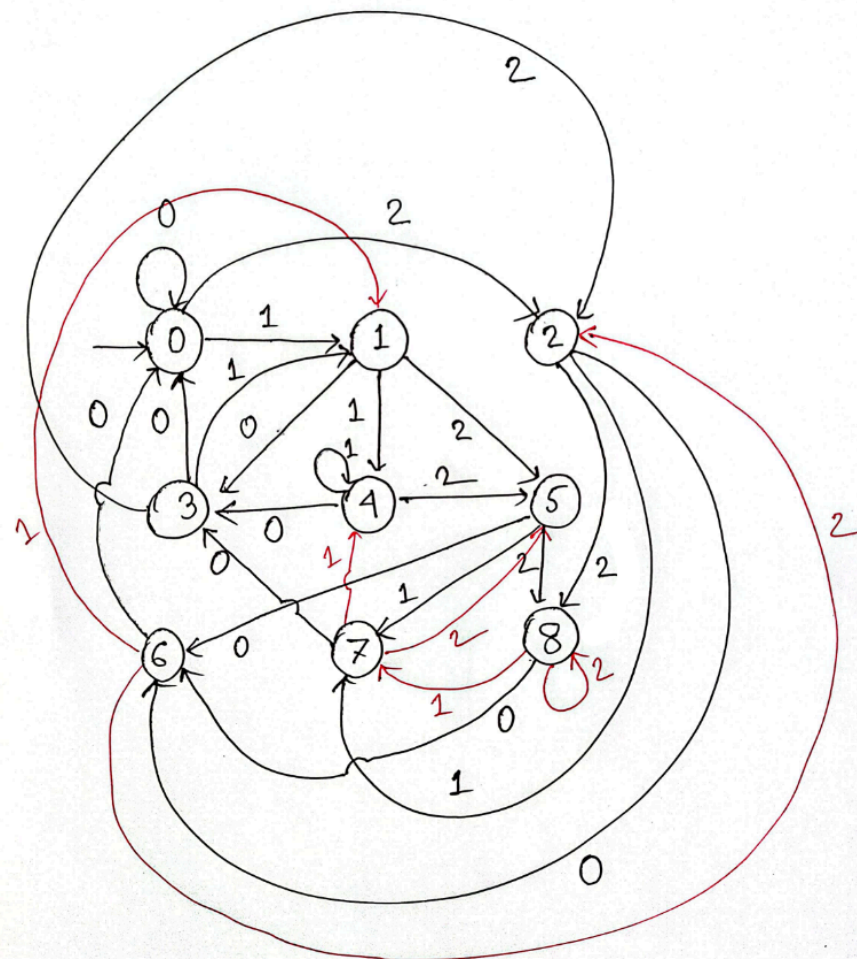
i)

In three base number system, the numbers which ends with 00, only those numbers are divisible by 9.

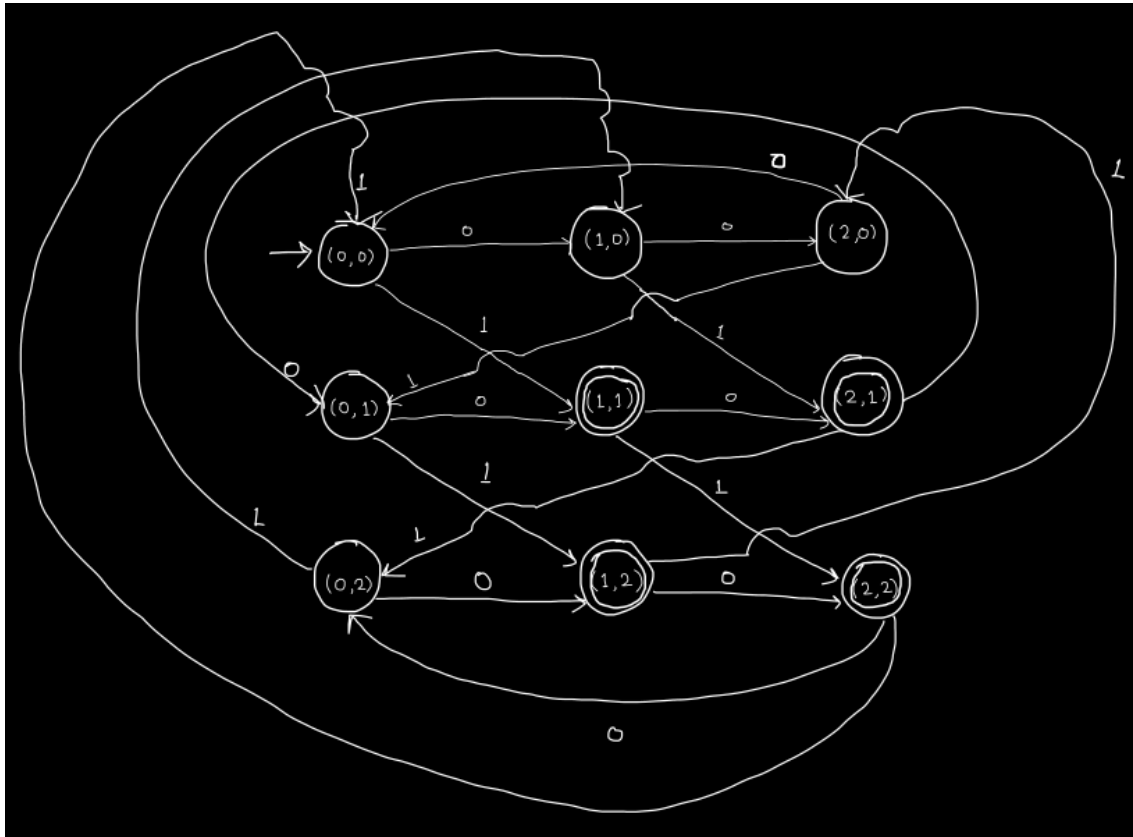


There is also a DFA with 9 states and 27 transitions - which is also correct.

The logic is - if there is a number x , in ternary representation, adding a 0 at the end will make the number $3x$, adding 1 at the end will make it $3x+1$ and for 2 it will become $3x+2$.



j)



A correct solution is anything equivalent to the above. The DFA diagram is a bit clunky due to there being a lot of states and transitions. However, the idea is simple: the states are indexed by a pair (i, j) where $0 \leq i, j \leq 2$, and i, j are the length and the number of 1s seen so far modulo 3. Upon seeing a new letter i goes up by 1, while j goes up by 1 only if the new letter is a 1.

The main observation is that $i^2 + j^2 = 2 \pmod 3$ iff $i = 1, 2$ and $j = 1, 2 \pmod 3$. This gives us four accepting states.

Any DFA for this language necessarily requires 9 states. So, any DFA with fewer states is wrong.

Problem 4

- a) Yes
- b) 1100
- c) 0100
- d) Yes
- e) No

CSE331 Section 07 & 08 Assignment 2 Solutions
Spring 2025

Question 1

a) $L1 = \{w \in \{0, 1\}^*: 0^i 1^j \text{ where } i \leq j\}$

Answer: Assume, for the sake of contradiction, $L1$ is regular. Then let p be the pumping length for $L1$. Now if we take the string: $w = 0^p 1^p \in L1$, then the length of w is $|w| = 2p \geq p$. So w can be split into xyz such that

1. $|xy| \leq p$,
2. $|y| > 0$ and
3. $xy^iz \in L1$ for each $i \geq 0$.

Assume, $x = 0^\alpha$, $y = 0^\beta$ and $z = 0^{p-\alpha-\beta} 1^p$

Since $|xy| \leq p$, and the first p characters of w are all 0's, we can conclude that y consists of only 0's.

Then, for $i = 2$, $xy^2z = 0^\alpha \cdot 0^{2\beta} \cdot 0^{p-\alpha-\beta} 1^p = 0^{p+\beta} 1^p \notin L1$ since β can not be zero ($|y| > 0$).

We have excess 0's in xy^2z . Thus we get a contradiction! Hence, $L1$ is not a regular language.

b) $L2 = \{w \in \{a, b, c\}^*: a^i b^j c^{k+2} \text{ where } i = k \text{ and } i, j, k \geq 0\}$

Answer: Assume, for the sake of contradiction, $L2$ is regular. Then let p be the pumping length for $L2$. Now if we take the string: $w = a^p b c^{p+2} \in L2$, then the length of w is $|w| = 2p+3 \geq p$. So w can be split into xyz such that

1. $|xy| \leq p$,
2. $|y| > 0$ and
3. $xy^iz \in L2$ for each $i \geq 0$.

Assume, $x = a^\alpha$, $y = a^\beta$ and $z = a^{p-\alpha-\beta} b c^{p+2}$

Since $|xy| \leq p$, and the first p characters of w are all a's, we can conclude that y consists of only a's.

Then, for $i = 2$, $xy^2z = a^\alpha \cdot a^{2\beta} \cdot a^{p-\alpha-\beta} b c^{p+2} = a^{p+\beta} b c^{p+2} \notin L2$ since β can not be zero ($|y| > 0$).

We have excess a's in xy^2z . Thus we get a contradiction! Hence, $L2$ is not a regular language.

c) $L3 = \{w1\#w2 \text{ such that } |w1| = 2 \cdot |w2|, \text{ where } \Sigma = \{0, 1\}\}$

Answer: Assume, for the sake of contradiction, $L3$ is regular. Then let p be the pumping length for $L3$. Now if we take the string: $w = 0^{2p} 1^p \in L3$, then the length of w is $|w| = 3p \geq p$. So w can be split into xyz such that

1. $|xy| \leq p$,
2. $|y| > 0$ and
3. $xy^iz \in L3$ for each $i \geq 0$.

Assume, $x = 0^\alpha$, $y = 0^\beta$ and $z = 0^{2p-\alpha-\beta} 1^p$

Since $|xy| \leq p$, and the first p characters of w are all 0's, we can conclude that y consists of only 0's.

Then, for $i = 2$, $xy^2z = 0^\alpha \cdot 0^{2\beta} \cdot 0^{2p-\alpha-\beta} 1^p = 0^{2p+\beta} 1^p \notin L3$ since β can not be zero ($|y| > 0$).

We have excess 0's in xy^2z . Thus we get a contradiction! Hence, $L3$ is not a regular language.

d) $L_4 = \{w \in \{a\}^*: a^{2^n} \text{ where } n \geq 0\}$

Answer: Assume, for the sake of contradiction, L_4 is regular. Then let p be the pumping length for L_4 . Now if we take the string: $w = a^{2^p} \in L_4$, then the length of w is $|w| = 2^p \geq p$. So w can be split into xyz such that

1. $|xy| \leq p$,
2. $|y| > 0$ and
3. $xy^iz \in L_4$ for each $i \geq 0$.

Assume, $x = a^\alpha$, $y = a^\beta$ and $z = a^{2^p - \alpha - \beta}$

Then, for $i = 2$, $xy^2z = a^\alpha \cdot a^{2\beta} \cdot a^{2^p - \alpha - \beta} = a^{2^p + \beta}$.

If $xy^2z \in L_4$, then $2^p + \beta$ should be a power of 2 (obviously $2^p + \beta > p$). As β can not be zero, the immediate next power of 2^p is 2^{p+1} .

So, $2^p + \beta > 2^{p+1}$

$\Rightarrow 2^p + \beta > 2 \cdot 2^p$

$\Rightarrow \beta > 2^p$

But from condition 1,

$|xy| \leq p$

$\Rightarrow |y| \leq p$

$\Rightarrow \beta \leq p$

Thus we get a contradiction! Hence, L_4 is not a regular language.

Question 2

A. $S \rightarrow 1P$

$P \rightarrow 0P \mid 1P \mid \epsilon$

Or, $S \rightarrow S0 \mid S1 \mid 1$

B. $S \rightarrow 0S0 \mid 1S1 \mid \epsilon$

C. $S \rightarrow P001P$

$P \rightarrow 0P \mid 1P \mid \epsilon$

D. $S \rightarrow 0S0 \mid 1$

E. $S \rightarrow PS1 \mid \epsilon$

$P \rightarrow 0 \mid \epsilon$

Or, $S \rightarrow 0S1 \mid Z$

$Z \rightarrow Z1 \mid \epsilon$

F. $S \rightarrow 1S1 \mid 0P$

$P \rightarrow 1P1 \mid 0$

G. $S \rightarrow 000S11 \mid X0X0X$

$X \rightarrow 0X \mid 1X \mid \epsilon$

Or, $S \rightarrow 000S11 \mid A0A0X$

$A \rightarrow 1A \mid \epsilon$

$X \rightarrow 0X \mid 1X \mid \epsilon$

H. $S \rightarrow R1P$

$R \rightarrow 000R \mid \epsilon$

$P \rightarrow 0P \mid \epsilon$

I. $0^i 10^j = 0^i 10^{2+3i} = 0^i 10^2 0^{3i} = 0^i \underline{1000} 0^{3i}$

$S \rightarrow 0S000 \mid 100$

$$\text{Or, } 0^i 1 0^j = 0^i 1 0^{2+3i} = 0^i 1 0^{3i} 0^2 = 0^i \underline{1} 0^{3i} \underline{00}$$

$$S \rightarrow P00$$

$$P \rightarrow 0P000 \mid 1$$

$$J. S \rightarrow P0R \mid Q$$

$$R \rightarrow 1111R \mid \epsilon$$

$$P \rightarrow 1P \mid \epsilon$$

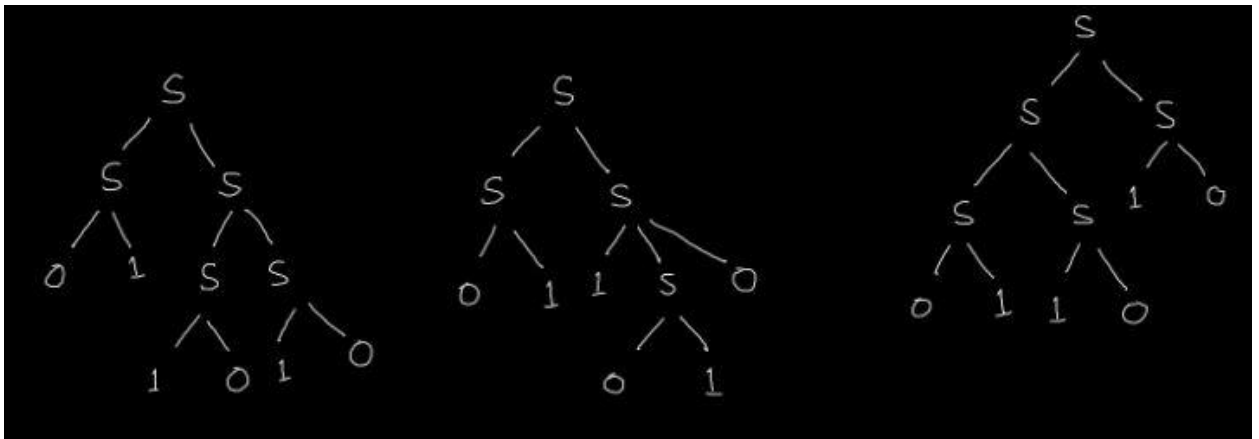
$$Q \rightarrow 11Q1 \mid 1110$$

Question 3

A.

a) There are three possible parse trees to derive the given string.

Drawing any two of the following will be correct.



b) Correct answers include 000111, 001110, 011100, 111000, 110001, 100011. Finding any two of these will be okay. Everything else is incorrect.

B.

a) & b)

There are three different derivations of abababb.

P (odd length palindrome)	E (even length substring)
a	bababb
aba	babb
ababa	bb

$$S \Rightarrow PE \Rightarrow aE \Rightarrow abaE \Rightarrow ababaE \Rightarrow abababbE \Rightarrow abababb$$
$$S \Rightarrow PE \Rightarrow aPaE \Rightarrow abaE \Rightarrow ababaE \Rightarrow abababbE \Rightarrow abababb$$
$$S \Rightarrow PE \Rightarrow aPaE \Rightarrow abPbaE \Rightarrow ababaE \Rightarrow abababbE \Rightarrow abababb$$

These are the three corresponding Parse Trees.

